COMP3153/9153
Algorithmic Verification

Lecture 12: Bounded Model Checking; SAT Solvers

Motivation

$$\mathcal{M} \models \varphi$$

Motivation

$$
\begin{array}{ccc}
\mathcal{M} & \models & \varphi \\
\downarrow & & \downarrow \\
A_{\mathcal{M}} & & A_{\varphi}
\end{array}
$$

Motivation

$$
\begin{array}{ccc}
\mathcal{M} & \models & \varphi \\
\downarrow & & \downarrow \\
L(A_{\mathcal{M}}) & \subseteq & L(A_{\varphi})
\end{array}
$$

Motivation

$$
\begin{tikzcd}
\mathcal{M} \arrow[d] & \models & \varphi \arrow[d] \\
L(A_{\mathcal{M}}) \arrow[dr] & \subseteq & L(A_{\varphi}) \arrow[dl] \\
& L(A) \stackrel{?}{=} \emptyset &
\end{tikzcd}
$$

Motivation

$$
\begin{tikzcd}[column sep=small]
\mathcal{M} \arrow[d] & \models & \varphi \arrow[d] \\
L(A_{\mathcal{M}}) \arrow[dr] & \subseteq & L(A_\varphi) \arrow[dl] \\
& L(A) \stackrel{?}{=} \emptyset &
\end{tikzcd}
$$

Difficulty

$A = A_{\mathcal{M}} \times A_{\neg\varphi}^{\text{local}} \times A_{\neg\varphi}^{\text{eventual}}$ will have $|\mathcal{M}| \cdot 2^{|\varphi|}$ states, so even linear-time algorithms can be impractical.

Motivation

$$
\begin{array}{ccc}
\mathcal{M} & \models & \varphi \\
\downarrow & & \downarrow \\
L(A_{\mathcal{M}}) & \subseteq & L(A_{\varphi}) \\
& \searrow \quad \swarrow & \\
& L(A) \stackrel{?}{=} \emptyset &
\end{array}
$$

Goal

Try to find a counterexample in k states, where $k = 1, 2, 3, \ldots$

Bounded Model Checking

Task

Given an automaton A, an LTL formula φ (in NNF), and a bound k.

Find a run ρ in A such that:

- $\rho \models \varphi$, and
- ρ uses at most $k+1$ states.

Bounded Model Checking

Task

Given an automaton A, an LTL formula φ (in NNF), and a bound k.

Find a run ρ in A such that:

- $\rho \models \varphi$, and
- ρ uses at most $k + 1$ states.

NB

This approach is also appropriate for "existential" LTL validity:

Say $A \models \mathbf{E}\varphi$ if there exists a run ρ in A such that $\rho \models \varphi$.

Bounded Model Checking

Task

Given an automaton A, an LTL formula φ (in NNF), and a bound k.

Find a run ρ in A such that:

- $\rho \models \varphi$, and
- ρ uses at most $k+1$ states.

NB

This approach is also appropriate for "existential" LTL validity:

Say $A \models \mathbf{E}\varphi$ if there exists a run ρ in A such that $\rho \models \varphi$.

NB

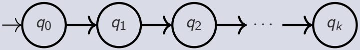
$$A \models \varphi \quad \Longleftrightarrow \quad A \not\models \mathbf{E}\neg\varphi$$

Bounded Semantics

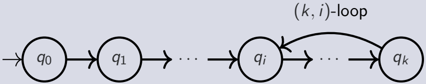
Bounded Models

We restrict our models to the subset of all paths in A with at most $k+1$ states. This gives two types of paths:

- Finite paths:



- Infinite (lasso) paths:



Bounded Semantics

Bounded Satisfaction

We will define a refinement of $\models$, $\models_k$, appropriate for bounded models.

We want **soundness**:

$$\rho \models_k \varphi \quad \Longrightarrow \quad \rho \models \varphi$$

Bounded Semantics

Bounded Satisfaction

We will define a refinement of $\models$, $\models_k$, appropriate for bounded models.

We want **soundness**:

$$\rho \models_k \varphi \quad \implies \quad \rho \models \varphi$$

If ρ is a lasso path, then we can define:

$$\rho \models_k \varphi \quad \iff \quad \rho \models \varphi$$

Bounded Semantics

Bounded Satisfaction

We will define a refinement of $\models$, $\models_k$, appropriate for bounded models.

We want **soundness**:

$$\rho \models_k \varphi \quad \Longrightarrow \quad \rho \models \varphi$$

If ρ is a lasso path, then we can define:

$$\rho \models_k \varphi \quad \Longleftrightarrow \quad \rho \models \varphi$$

Question

How to deal with finite paths?

Bounded Satisfaction

Definition

For $0 \leq i \leq k$ define $\models_k^i$ as follows.
For finite path ρ, and LTL formula φ (in NNF):

Bounded Satisfaction

Definition

For $0 \leq i \leq k$ define $\models^i_k$ as follows.
For finite path ρ, and LTL formula φ (in NNF):

- $\varphi = p$: $\rho \models^i_k \varphi$ if $p \in L(\rho_i)$
- $\varphi = \psi_1 \wedge \psi_2$: $\rho \models^i_k \varphi$ if $\rho \models^i_k \psi_1$ and $\rho \models^i_k \psi_2$

Bounded Satisfaction

Definition

For $0 \leq i \leq k$ define $\models^i_k$ as follows.
For finite path ρ, and LTL formula φ (in NNF):

- $\varphi = \neg p$: $\rho \models^i_k \varphi$ if $p \notin L(\rho_i)$
- $\varphi = \psi_1 \vee \psi_2$: $\rho \models^i_k \varphi$ if $\rho \models^i_k \psi_1$ or $\rho \models^i_k \psi_2$

Bounded Satisfaction

Definition

For $0 \leq i \leq k$ define $\models_k^i$ as follows.
For finite path ρ, and LTL formula φ (in NNF):

- $\varphi = p$: $\rho \models_k^i \varphi$ if $p \in L(\rho_i)$
- $\varphi = \psi_1 \wedge \psi_2$: $\rho \models_k^i \varphi$ if $\rho \models_k^i \psi_1$ and $\rho \models_k^i \psi_2$
- $\varphi = \mathbf{X}\psi$: For $0 \leq i < k$, $\rho \models_k^i \varphi$ if $\rho \models_k^{i+1} \psi$

Bounded Satisfaction

Definition

For $0 \leq i \leq k$ define $\models_k^i$ as follows.
For finite path ρ, and LTL formula φ (in NNF):

- $\varphi = p$: $\rho \models_k^i \varphi$ if $p \in L(\rho_i)$
- $\varphi = \psi_1 \wedge \psi_2$: $\rho \models_k^i \varphi$ if $\rho \models_k^i \psi_1$ and $\rho \models_k^i \psi_2$
- $\varphi = \mathbf{X}\psi$: For $0 \leq i < k$, $\rho \models_k^i \varphi$ if $\rho \models_k^{i+1} \psi$
- $\varphi = \psi_1 \mathbf{U} \psi_2$: $\rho \models_k^i \varphi$ if there is a j, $i \leq j \leq k$ such that $\rho \models_k^j \psi_2$ and $\rho \models_k^m \psi_1$ for all $m \in [i, j)$

Bounded Satisfaction

Definition

For $0 \leq i \leq k$ define $\models^i_k$ as follows.
For finite path ρ, and LTL formula φ (in NNF):

- $\varphi = p$: $\rho \models^i_k \varphi$ if $p \in L(\rho_i)$
- $\varphi = \psi_1 \wedge \psi_2$: $\rho \models^i_k \varphi$ if $\rho \models^i_k \psi_1$ and $\rho \models^i_k \psi_2$
- $\varphi = \mathbf{X}\psi$: For $0 \leq i < k$, $\rho \models^i_k \varphi$ if $\rho \models^{i+1}_k \psi$
- $\varphi = \psi_1 \, \mathbf{U} \, \psi_2$: $\rho \models^i_k \varphi$ if there is a j, $i \leq j \leq k$ such that $\rho \models^j_k \psi_2$ and $\rho \models^m_k \psi_1$ for all $m \in [i, j)$
- $\varphi = \psi_1 \mathbf{R} \psi_2$: $\rho \models^i_k \varphi$ if there is a j, $i \leq j \leq k$ such that $\rho \models^j_k \psi_1$ and $\rho \models^m_k \psi_2$ for all $m \in [i, j]$.

Bounded Satisfaction

Definition

For $0 \leq i \leq k$ define $\models^i_k$ as follows.
For finite path ρ, and LTL formula φ (in NNF):

- $\varphi = p$: $\rho \models^i_k \varphi$ if $p \in L(\rho_i)$
- $\varphi = \psi_1 \wedge \psi_2$: $\rho \models^i_k \varphi$ if $\rho \models^i_k \psi_1$ and $\rho \models^i_k \psi_2$
- $\varphi = \mathbf{X}\psi$: For $0 \leq i < k$, $\rho \models^i_k \varphi$ if $\rho \models^{i+1}_k \psi$
- $\varphi = \psi_1 \mathbf{U} \psi_2$: $\rho \models^i_k \varphi$ if there is a j, $i \leq j \leq k$ such that $\rho \models^j_k \psi_2$ and $\rho \models^m_k \psi_1$ for all $m \in [i, j)$
- $\varphi = \psi_1 \mathbf{R} \psi_2$: $\rho \models^i_k \varphi$ if there is a j, $i \leq j \leq k$ such that $\rho \models^j_k \psi_1$ and $\rho \models^m_k \psi_2$ for all $m \in [i, j]$.

NB

It is never the case that $\rho \models^i_k \mathbf{G}\varphi$.

Bounded Satisfaction

Definition

For $0 \leq i \leq k$ define $\models^i_k$ as follows.
For finite path ρ, and LTL formula φ (in NNF):

- $\varphi = p$: $\rho \models^i_k \varphi$ if $p \in L(\rho_i)$
- $\varphi = \psi_1 \wedge \psi_2$: $\rho \models^i_k \varphi$ if $\rho \models^i_k \psi_1$ and $\rho \models^i_k \psi_2$
- $\varphi = \mathbf{X}\psi$: For $0 \leq i < k$, $\rho \models^i_k \varphi$ if $\rho \models^{i+1}_k \psi$
- $\varphi = \psi_1 \,\mathbf{U}\, \psi_2$: $\rho \models^i_k \varphi$ if there is a j, $i \leq j \leq k$ such that $\rho \models^j_k \psi_2$ and $\rho \models^m_k \psi_1$ for all $m \in [i, j)$
- $\varphi = \psi_1 \mathbf{R} \psi_2$: $\rho \models^i_k \varphi$ if there is a j, $i \leq j \leq k$ such that $\rho \models^j_k \psi_1$ and $\rho \models^m_k \psi_2$ for all $m \in [i, j]$.

We say $\rho \models_k \varphi$ if $\rho \models^0_k \varphi$.

NB

It is never the case that $\rho \models^i_k \mathbf{G}\varphi$.

Bounded Semantics: Questions

Question

In ordinary LTL semantics:

$$\rho \models \mathbf{G}\varphi \quad \Longleftrightarrow \quad \rho \models \neg\mathbf{F}\neg\varphi$$

Is it true that:

$$\rho \models_k \mathbf{G}\varphi \quad \Longleftrightarrow \quad \rho \models_k \neg\mathbf{F}\neg\varphi?$$

Completeness

Question

We have, for all k:

$$\rho \models_k \varphi \quad \Longrightarrow \quad \rho \models \varphi$$

so

$$A \models_k \mathbf{E}\varphi \quad \Longrightarrow \quad A \models \mathbf{E}\varphi.$$

When does the converse hold?

Completeness

Question

We have, for all k:

$$\rho \models_k \varphi \quad \Longrightarrow \quad \rho \models \varphi$$

so

$$A \models_k \mathbf{E}\varphi \quad \Longrightarrow \quad A \models \mathbf{E}\varphi.$$

When does the converse hold?

By the construction of an LTL acceptance automaton, we know for $k = |A| \cdot 2^{|\varphi|}$:

$$A \models \mathbf{E}\varphi \quad \Longrightarrow \quad A \models_k \mathbf{E}\varphi.$$

Can we improve on this?

Completeness

Question

We have, for all k:

$$\rho \models_k \varphi \quad \Longrightarrow \quad \rho \models \varphi$$

so

$$A \models_k \mathbf{E}\varphi \quad \Longrightarrow \quad A \models \mathbf{E}\varphi.$$

When does the converse hold?

By the construction of an LTL acceptance automaton, we know for $k = |A| \cdot 2^{|\varphi|}$:

$$A \models \mathbf{E}\varphi \quad \Longrightarrow \quad A \models_k \mathbf{E}\varphi.$$

Can we improve on this?

For some formulas, yes. For example for $\varphi = \mathbf{EF}p$ we need only consider the *diameter* of A (longest shortest path between nodes)

Model Checking with SAT Solvers

Goal

Given a model A and an LTL formula φ, define CNF formulas $\mathrm{For}(A)$ and $\mathrm{For}(\varphi)_k$ so that:

$$A \models_k \mathbf{E}\varphi \quad \Longleftrightarrow \mathrm{For}(A) \land \mathrm{For}(\varphi)_k \text{ is satisfiable}$$

Model Checking with SAT Solvers

Goal

Given a model A and an LTL formula φ, define CNF formulas $\mathsf{For}(A)$ and $\mathsf{For}(\varphi)_k$ so that:

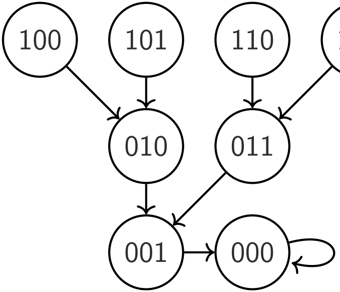
$$A \models_k \mathbf{E}\varphi \iff \mathsf{For}(A) \wedge \mathsf{For}(\varphi)_k \text{ is satisfiable}$$

Why?

The development of SAT solvers which can efficiently solve multi-million variable instances (especially those obtained from real-world applications)

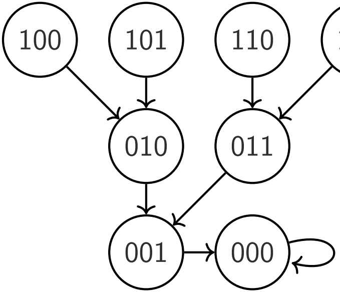
Motivating example

Consider the 3-bit (Right) Shift Register:



Motivating example

Consider the 3-bit (Right) Shift Register:

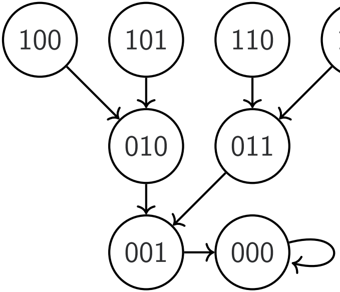


Store register value in vector $\mathbf{x}$. Encode the transition relation with next state $\mathbf{x}'$

Consider the requirement: $\varphi = \mathbf{F}(\mathbf{x} = 0)$

Motivating example

Consider the 3-bit (Right) Shift Register:



Store register value in vector $\mathbf{x}$. Encode the transition relation with next state $\mathbf{x}'$

Consider the requirement: $\varphi = \mathbf{F}(\mathbf{x} = 0)$

Question

What are the corresponding formulas $\mathrm{For}(A)$ and $\mathrm{For}(\varphi)_3$?

Using SAT solvers

See supplementary slides.

Weekly Feedback

I would appreciate any comments/suggestions/requests you have on this week's lectures (and tutorials).

https://forms.office.com/r/sCi2xjavrS